

DILLAN IMANS

Medical AI Researcher

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SUMMARY

Medical AI Researcher with first-author publications at EMA4MICCAI, IEEE BIBM, and Diagnostics, working on data- and label-efficient methods, including knowledge distillation, unsupervised domain adaptation, and unpaired cross-modal learning for diagnostic imaging and surgical vision.

EDUCATION

B.S. Computer Science & Engineering, Sungkyunkwan University Aug 2022 - Aug 2026
95.7 / 100 · Dean's List 2023 · Academic Excellence Scholarship

EXPERIENCE

AI Research Engineer · Superintelligence Lab SKKU, South Korea Oct 2024 - Jun 2026

- Developed unsupervised domain adaptation framework for brain tumor segmentation across MRI domain shift, outperforming benchmarks by ~5–10% AUC (First Author, IEEE BIBM 2025)
- Built end-to-end retinal analysis system covering optic disc/cup segmentation, vessel analysis, and glaucoma detection; optimized for clinical deployment via ONNX with low-latency inference, integrated with industry-partnered screening platform.
- Designed graph-based distillation framework for unpaired MRI-to-fundus hypertension prediction, outperforming standard distillation baselines (First Author, EMA4MICCAI 2026)
- Designed a codebook-based distillation framework for unpaired medical image classification, improving student accuracy by 2.5–5.7 macro-F1 across two clinical tasks (First Author, preprint under review at ACCV 2026)

AI Researcher · Labren CUHK, Hong Kong Jun 2024 - Jun 2026

- Designed preprocessing infrastructure and annotation QA pipeline for the first large-scale surgical action recognition dataset (11,915 clips)
- Curated a dataset and ran initial model experiments for a multimodal surgical robotics paper in collaboration with NVIDIA, contributing as a co-author
- Co-authored 3 manuscripts in submission at npj Digital Surgery and top-tier conferences

AI Researcher · Infolab SKKU, South Korea Feb 2024 - Oct 2024

- Developed explainable ensemble framework for depression detection and severity prediction, benchmarking classical, ensemble, and dynamic selection algorithms, achieving 88.3% accuracy across 2,763 clinical records with SHAP interpretability (First Author, Diagnostics 2024)

PUBLICATIONS

Full list: scholar.google.com/citations?user=o-TtUq0AAAAJ

- Imans D. et al. "Clinical Graph-Mediated Distillation for Unpaired MRI-to-CFI Hypertension Prediction." EMA4MICCAI 2026.
- Imans D. et al. "Unsupervised Domain Adaptation with SAM-RefiSeR for Enhanced Brain Tumor Segmentation." IEEE BIBM 2025.
- Imans D. et al. "Explainable Multi-Layer Dynamic Ensemble Framework Optimized for Depression Detection and Severity Assessment." Diagnostics, 14(21), 2385 (2024).
- 1 first-author preprint on arXiv · 3 co-authored preprints under review

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java

ML / AI: PyTorch, MONAI, HuggingFace, scikit-learn, OpenCV, wandb, TensorBoard

Medical / Bio: NiBabel, SimpleITK, pydicom

Tools: Git, Linux, Docker

Spoken Languages: English & Indonesian (Fluent), Mandarin & Korean (Conversational)